

ASHUTOSH ROY

AI/ML Specialist — Vision & RAG

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SUMMARY

AI/ML Specialist specializing in Computer Vision (OCR, detection, classification) and Retrieval-Augmented Generation (RAG) architectures. Published researcher with a track record in building high-accuracy Vision-RAG systems — from benchmarking OCR models at IIT Delhi to architecting multimodal retrieval pipelines. Highly proficient in Python, PyTorch/TensorFlow, and vector databases (FAISS, Pinecone), with a focus on data-driven model evaluation and intelligent search solutions.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Evaluated and benchmarked 3 OCR systems (Tesseract, Marker, Gemini OCR) for image classification and text detection on diverse historical manuscripts, optimizing extraction fidelity.
- Designed a per-class model selection strategy to reduce character error rates, integrating vision-based document parsing into a scalable archival pipeline.
- Built a multimodal RAG retrieval pipeline combining semantic vector search (FAISS/Pinecone) and keyword search (Apache Solr) over thousands of archival images.
- Fine-tuned **Surya OCR** and engineered custom data preprocessing/annotation pipelines to isolate semantic data from scanned historical books and maps.

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Built an OCR-based analysis module for image detection and text extraction from complex medical reports, ensuring high accuracy via custom OpenCV preprocessing.
- Developed end-to-end Vision-NLP pipelines, integrating Tesseract OCR with LLM architectures for real-time automated data annotation and analysis.
- Optimized model inference latency by 30% through efficient batch processing and quantization, improving performance of AI-driven healthcare applications.

PROJECTS

Speech Emotion & Stress Detection | Python, PyTorch, Wav2Vec2, BiLSTM, Hugging Face

Jan 2026 – May 2026

- Engineered a comprehensive audio preprocessing pipeline integrating Voice Activity Detection (VAD), silence removal, log-Mel spectrogram generation, and MFCC extraction.
- Unified and standardized data across 3 distinct emotional speech corpora (RAVDESS, TESS, SAVEE) to create a robust training dataset.
- Developed a novel hybrid **Wav2Vec2 + BiLSTM** architecture that effectively captures both spatial acoustic features and temporal sequence dependencies.
- Achieved state-of-the-art **85.9% accuracy**, **0.856 F1 score**, and an RMSE of **0.1268** on continuous stress regression, significantly outperforming traditional SVM, Random Forest, and standalone CNN baselines.

Exam-Helper RAG System | Python, LLaMA 3, FAISS, LangChain, Streamlit

Dec 2024 – Apr 2025

- Implemented a robust multimodal document ingestion framework supporting complex scanned PDFs, handwritten notes, and image-based Q&A workflows.
- Combined advanced vision-based document parsing with dense vector retrieval using FAISS to ensure highly context-accurate and hallucination-free LLM responses.
- Integrated LLaMA 3 via LangChain for nuanced conversational abilities, wrapped in an intuitive Streamlit user interface for seamless student interaction.

Music Identification App | Python, ACRCLOUD API, Streamlit, NumPy

Sep 2024 – Dec 2024

- Built a real-time audio signal processing pipeline encompassing microphone waveform capture, digital filtering, and localized feature extraction.
- Implemented highly accurate acoustic fingerprinting and metadata retrieval via integration with the ACRCLOUD API.
- Designed and deployed a responsive, cross-platform user interface using Flutter, emphasizing low-latency interactions and aesthetic visual feedback.

Medical Chatbot with OCR Pipeline | Python, Mistral 7B, QLoRA, Tesseract, OpenCV

Mar 2024 – Jun 2024

- Architected an advanced document vision pipeline utilizing Tesseract OCR coupled with robust **OpenCV** preprocessing techniques, including morphological denoising, adaptive binarization, and automated deskewing.
- Extracted and mapped complex structured text from varied clinical report images into standardized JSON formats for downstream processing.
- Applied QLoRA techniques to fine-tune a Mistral 7B model specifically for medical Natural Language Understanding (NLU), resulting in a highly specialized, context-aware healthcare chatbot.

PUBLICATIONS

SARCH: Multimodal Search for Archaeological Archives | *Co-author*

Nov 2025

- Co-authored research detailing a comprehensive multimodal search pipeline for digitizing and indexing extensive historical archives.
- Successfully combined advanced OCR techniques, convolutional image classification, and dense vector hybrid search to enable highly scalable and accurate retrieval of archival data.

Speech Emotion & Stress Detection using Hybrid Architecture | *First Author*

May 2026

- First-authored paper detailing the design, implementation, and evaluation of a novel Wav2Vec2 + BiLSTM framework for continuous stress regression.
- Demonstrated state-of-the-art performance with an 85.9% accuracy rate, providing deep insights into feature extraction for affective computing.
- Peer-reviewed and accepted for presentation at the prestigious **RECCAP 2026** conference hosted at IIT Palakkad.

TECHNICAL SKILLS

Computer Vision: OCR (Tesseract, Surya, Gemini OCR), Image Detection/Classification, Document Understanding, Image Preprocessing, OpenCV

RAG & LLMs: Retrieval-Augmented Generation, Dense Retrieval, Vector Databases (FAISS, Pinecone), Prompt Engineering, Transformer Fine-Tuning

Machine Learning: Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face, Data Annotation, Evaluation & Benchmarking

Libraries & Search: Pandas, NumPy, Apache Solr, FastAPI, LangChain, Streamlit

Tools: Git, Docker, Linux, Jupyter Notebook, Google Colab, VS Code

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

- Relevant Coursework: Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Advanced Mathematics for Machine Learning.
- Active participant in AI/ML research groups, consistently leading complex technical projects and academic publications.

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana

- Specialized in Physics, Chemistry, Mathematics, and Biology (PCMB), developing a strong foundational understanding of analytical problem-solving.